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Vahan.ai — Product Analytics & ML Case Study

End-to-End Pipeline for Candidate Conversion Optimization

Presenter: Jegadeesh D

Python | PostgreSQL | Streamlit | Scikit-Learn

The Challenge: Extreme Class Imbalance

18.2K

Leads

18,198 leads across 16 cohorts and 11 upload dates

54

FT Trips

Only 54 candidates reached "First Trip" milestone

0.30%

Conversion

Extremely low positive conversion rate

The Bottleneck

Only 54 candidates out of the entire pool successfully reached the **"First Trip" (FT)** milestone, representing a severe operational drop-off.

Operational Impact

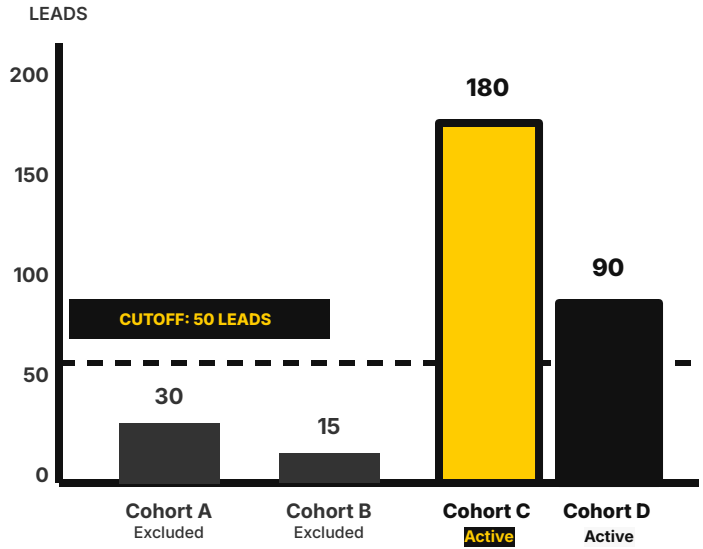
A **~0.30% positive conversion rate** demands early prediction of high-intent candidates. Efficient targeting is required to drastically reduce wasted tele-calling effort and streamline lead pipelines.

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Cohort Performance Ranking

Focusing on monetization-critical cohorts through rigorous volume filtering

Cohort Lead Volume



NORTH STAR METRIC

FT Conversion Rate

BUSINESS RATIONALE

Monetization depends entirely on completed first trips, not upstream metrics like connect or interest rates.

STATISTICAL RIGOR

Excluded cohorts with under 50 leads to eliminate noise and false positives.

Database Engineering & Aggregation

Execution & Grain

- **Native PostgreSQL Execution**

  Executed directly against the local database to optimize pipeline performance.

- **Cohort-Date Level Grain**

  Aggregated at the strict combination level of upload_date × lead_source.

Data Integrity & Precision

- **Zero-Division Avoidance**

  Handled division-by-zero safely using explicit NULLIF() checks in SQL operations.

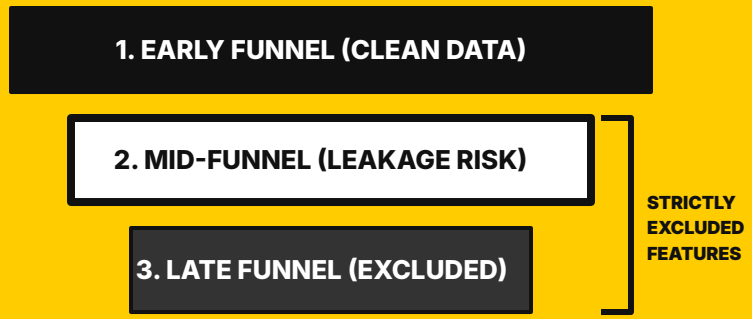
- **Explicit Numeric Casting**

  Ensured absolute calculations precision via explicit numeric type casting for averages.

Predictive ML & Addressing Leakage

THE LEAKAGE PROBLEM

Mid-funnel events occur post-connection, skewing historical ML model predictions.



Clean Modeling & Evaluation

PRODUCTION PIPELINE

MODELS EVALUATED

Logistic Regression, Random Forest, & XGBoost

THE LEAKAGE SOLUTION

Strictly excluded late-stage features. Scoring targets freshly uploaded leads using only early-funnel data to guarantee realistic live performance.

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Model Selection for Production

01 / CHOSEN MODEL

Random Forest was selected as the final production model due to its robust prediction stability and optimal fit.

02 / OPERATIONAL DRIVERS

Model decisions are heavily driven by key operational features: **time to first attempt** , **lead source** , and **attempt frequency** .

03 / TEAM USABILITY

The implementation provides immediate, actionable levers for **tele-calling teams** to dynamically adjust outreach strategies.

04 / OPTIMIZATION

Hyperparameters and model performance were rigorously tuned and optimized via a structured **3-fold GridSearchCV** process.

Engineering & Deployment Architecture

WEB

INTERFACE

Interactive, multi-page Streamlit web dashboard.

17

RELIABILITY

Backed by 17 automated Pytest unit tests.

SHA

SECURITY

Dynamic credential parsing and SHA-256 hashing to protect candidate phone numbers.

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Conclusion & Next Steps

Business Impact

Empowers teams to prioritize high-intent leads, eliminating wasted effort on the 99.7% non-converting segment.

Next Steps

Launch live A/B testing with the tele-calling floor to measure the actual lift in FT Conversion.

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